BLITZREMESHER

VEGETATION RESEARCH / MEASURED COMPARISONS
28 SEPTEMBER 2026

C++20 MESH REDUCTION · GAME-ENGINE IMPORT PIPELINES

Fewer triangles.
Measured transitions.

Six nature assets in both vertex modes. Fern and Broadleaf tree have paired before/after chains. Every tile contains the actual baked geometry; use target scale to inspect the distant LODs.

Geometry, with an error budget.

Propose a simpler mesh. Check it against the previous level and the original. Keep the smallest valid chain found.

> The new comparisons use the same eight-proposal budget and sampled pixel gates. Dense tail checks are reported separately and can fail. Coverage change is a filled-geometry diagnostic, not an opacity or shading score. Rows 01–04 are archived opaque examples.

SHARED · TAIL IMPROVEMENT	REBUILT · TAIL IMPROVEMENT	NATURE RESEARCH COLLECTION	DENSE TAIL CHECKS PASSED
55.8%	49.4%	40	16/16
Lower retained ratio vs control · same six assets, category balance and budget	Lower retained ratio vs control · same six assets, category balance and budget	28 development models · 12 validation models · families kept together	706-camera set · source and adjacent checks · 8× to 32× sampling

01 / ACTUAL OUTPUT GEOMETRY

From source to 16 pixels.

Each tile is an actual exported LOD mesh. Nature rows audit full card geometry; texture opacity is not scored.

MATCHED COMPARISON	FINAL TRIANGLES	LAST THREE TRIANGLES	BAKE SECONDS	FINAL COVERAGE CHANGED
Grass clump shared vertices · 8 proposals	1,008 → **382**	4,032 → 2,677	2.31 → 13.76	16.8% → 26.3%
Grass clump rebuilt vertices · 8 proposals	1,008 → **245**	4,032 → 2,539	2.30 → 15.42	15.6% → 35.5%
Fern fronds shared vertices · 8 proposals	482 → **260**	1,929 → 780	3.82 → 31.63	14.4% → 21.1%
Fern fronds rebuilt vertices · 8 proposals	48 → **16**	1,495 → 178	3.75 → 31.26	42.5% → 59.1%
Leafy shrub shared vertices · 8 proposals	343 → **244**	2,647 → 2,404	6.63 → 28.89	77.9% → 78.6%
Leafy shrub rebuilt vertices · 8 proposals	94 → **136**	2,398 → 1,765	6.25 → 31.19	73.8% → 76.3%
Broadleaf tree shared vertices · 8 proposals	523 → **110**	2,176 → 770	0.80 → 8.30	23.7% → 40.2%
Broadleaf tree rebuilt vertices · 8 proposals	704 → **110**	2,357 → 1,137	0.73 → 9.91	4.2% → 29.7%
Flowering ground cover shared vertices · 8 proposals	356 → **836**	11,042 → 5,845	7.05 → 28.10	54.4% → 42.2%
Flowering ground cover rebuilt vertices · 8 proposals	356 → **664**	11,042 → 6,853	13.89 → 31.91	57.3% → 47.3%
Leafy branch shared vertices · 8 proposals	48 → **10**	253 → 73	0.71 → 4.08	44.2% → 54.4%
Leafy branch rebuilt vertices · 8 proposals	31 → **10**	155 → 74	0.93 → 8.35	60.0% → 52.4%

DISPLAY **Clay** Wireframe Silhouette · LEVELS **Audit slots** Runtime levels · SCALE **Enlarged** Target px · Reset views ↺

Enlarged for inspection. Same camera and source scale across each row. Drag a mesh to rotate its chain; click a tile to inspect it.

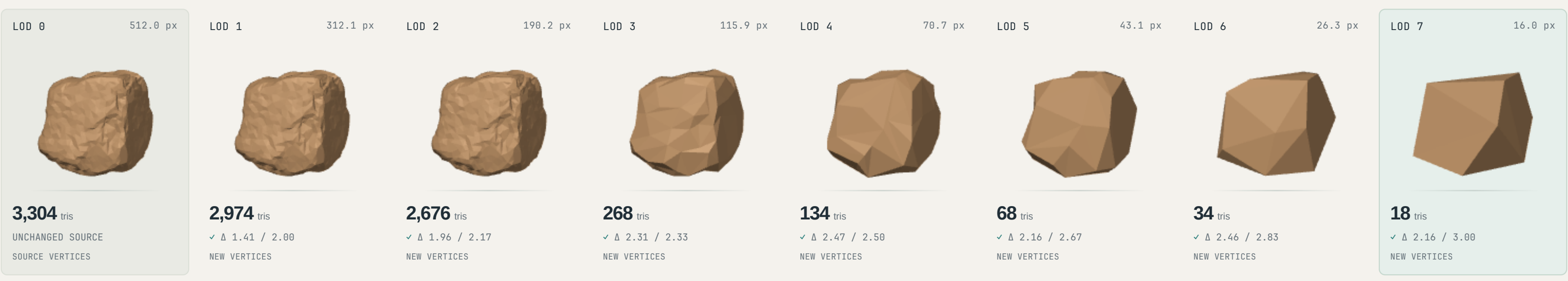
01 / ROCK

Moon rock

Rock surface detail gives way to the permitted distant silhouette. · 8 runtime levels / 8 audit slots

Bake pipeline: 48.80 s · **Vertices: rebuilt / new buffers** · time includes import, audit and export

−99.5% at LOD 7
3,304 → 18 tris



LOD 0 512.0 px	LOD 1 312.1 px	LOD 2 190.2 px	LOD 3 115.9 px	LOD 4 70.7 px	LOD 5 43.1 px	LOD 6 26.3 px	LOD 7 16.0 px
3,304 tris	2,974 tris	2,676 tris	268 tris	134 tris	68 tris	34 tris	18 tris
UNCHANGED SOURCE	✓ Δ 1.41 / 2.00	✓ Δ 1.96 / 2.17	✓ Δ 2.31 / 2.33	✓ Δ 2.47 / 2.50	✓ Δ 2.16 / 2.67	✓ Δ 2.46 / 2.83	✓ Δ 2.16 / 3.00
SOURCE VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES

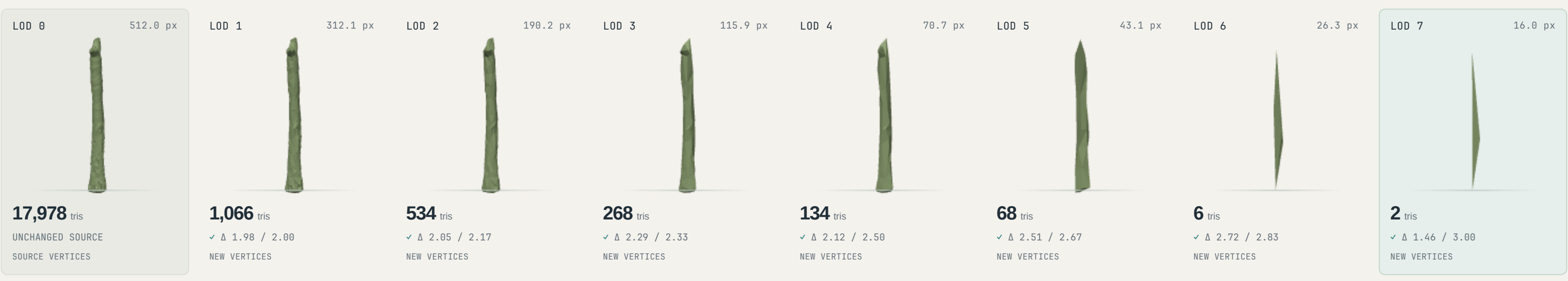
02 / ORGANIC

Quiver tree trunk

The extreme tail passes pixel-distance gates but loses filled area. · 8 runtime levels / 8 audit slots

Bake pipeline: 75.89 s · **Vertices: rebuilt / new buffers** · time includes import, audit and export

−99.99% at LOD 7
17,978 → 2 tris



LOD 0 512.0 px	LOD 1 312.1 px	LOD 2 190.2 px	LOD 3 115.9 px	LOD 4 70.7 px	LOD 5 43.1 px	LOD 6 26.3 px	LOD 7 16.0 px
17,978 tris	1,066 tris	534 tris	268 tris	134 tris	68 tris	6 tris	2 tris
UNCHANGED SOURCE	✓ Δ 1.98 / 2.00	✓ Δ 2.05 / 2.17	✓ Δ 2.29 / 2.33	✓ Δ 2.12 / 2.50	✓ Δ 2.51 / 2.67	✓ Δ 2.72 / 2.83	✓ Δ 1.46 / 3.00
SOURCE VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES

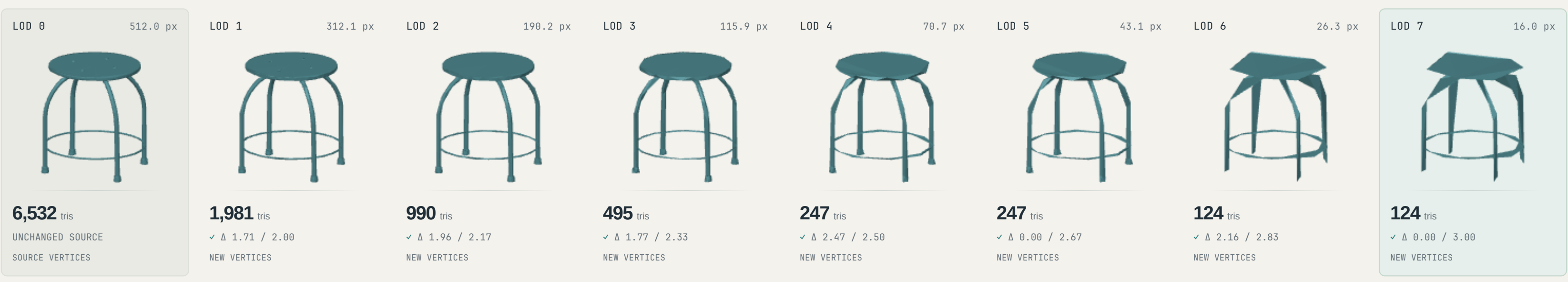
03 / MANUFACTURED

Metal stool

This asset still needs better intermediate proposals. · 6 runtime levels / 8 audit slots

Bake pipeline: 45.19 s · **Vertices: rebuilt / new buffers** · time includes import, audit and export

−98.1% at LOD 7
6,532 → 124 tris



LOD 0 512.0 px	LOD 1 312.1 px	LOD 2 190.2 px	LOD 3 115.9 px	LOD 4 70.7 px	LOD 5 43.1 px	LOD 6 26.3 px	LOD 7 16.0 px
6,532 tris	1,981 tris	990 tris	495 tris	247 tris	247 tris	124 tris	124 tris
UNCHANGED SOURCE	✓ Δ 1.71 / 2.00	✓ Δ 1.96 / 2.17	✓ Δ 1.77 / 2.33	✓ Δ 2.47 / 2.50	✓ Δ 0.00 / 2.67	✓ Δ 2.16 / 2.83	✓ Δ 0.00 / 3.00
SOURCE VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES

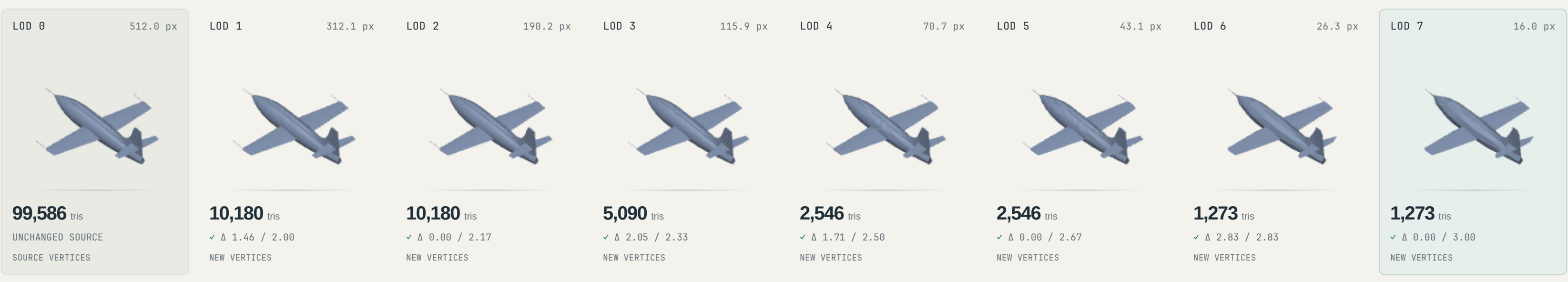
04 / COMPLEX SCAN

Bell X-1

Repeated audit slots can share one runtime mesh. · 5 runtime levels / 8 audit slots

Bake pipeline: 57.71 s · **Vertices: rebuilt / new buffers** · time includes import, audit and export

−98.7% at LOD 7
99,586 → 1,273 tris



LOD 0 512.0 px	LOD 1 312.1 px	LOD 2 190.2 px	LOD 3 115.9 px	LOD 4 70.7 px	LOD 5 43.1 px	LOD 6 26.3 px	LOD 7 16.0 px
99,586 tris	10,180 tris	10,180 tris	5,090 tris	2,546 tris	2,546 tris	1,273 tris	1,273 tris
UNCHANGED SOURCE	✓ Δ 1.46 / 2.00	✓ Δ 0.00 / 2.17	✓ Δ 2.05 / 2.33	✓ Δ 1.71 / 2.50	✓ Δ 0.00 / 2.67	✓ Δ 2.83 / 2.83	✓ Δ 0.00 / 3.00
SOURCE VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES

05 / NATURE · FERN

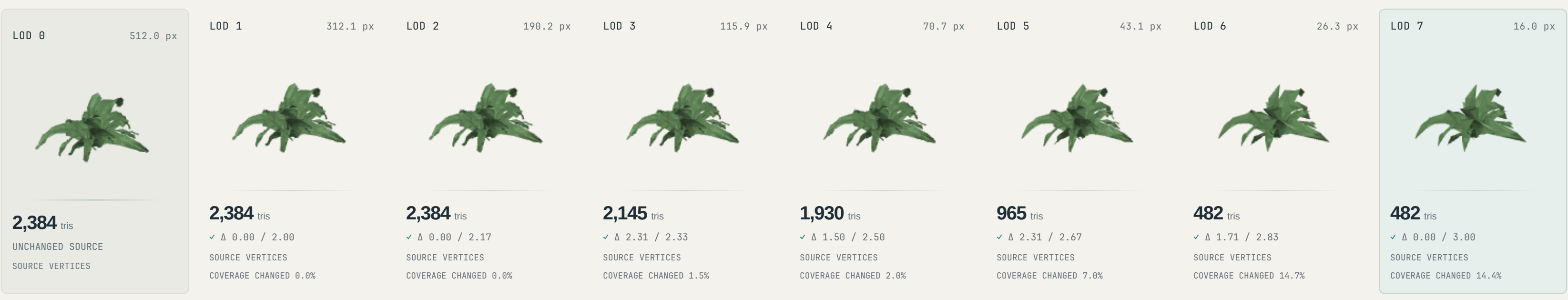
Fern fronds · before / shared

8 proposals per level; identical pixel limits. Geometry-only, without opacity or shading. · 5 runtime levels / 8 audit slots

Chain bake: 3.82 s · **Vertices: shared original buffer** · generation + audit

Dense tail check: passed (706 views)

−79.8% at LOD 7
2,384 → 482 tris



LOD 0 512.0 px	LOD 1 312.1 px	LOD 2 190.2 px	LOD 3 115.9 px	LOD 4 70.7 px	LOD 5 43.1 px	LOD 6 26.3 px	LOD 7 16.0 px
2,384 tris	2,384 tris	2,384 tris	2,145 tris	1,930 tris	965 tris	482 tris	482 tris
UNCHANGED SOURCE	✓ Δ 0.00 / 2.00	✓ Δ 0.00 / 2.17	✓ Δ 2.31 / 2.33	✓ Δ 1.50 / 2.50	✓ Δ 2.31 / 2.67	✓ Δ 1.71 / 2.83	✓ Δ 0.00 / 3.00
SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES
	COVERAGE CHANGED 0.0%	COVERAGE CHANGED 0.0%	COVERAGE CHANGED 1.5%	COVERAGE CHANGED 2.0%	COVERAGE CHANGED 7.0%	COVERAGE CHANGED 14.7%	COVERAGE CHANGED 14.4%

06 / NATURE · FERN

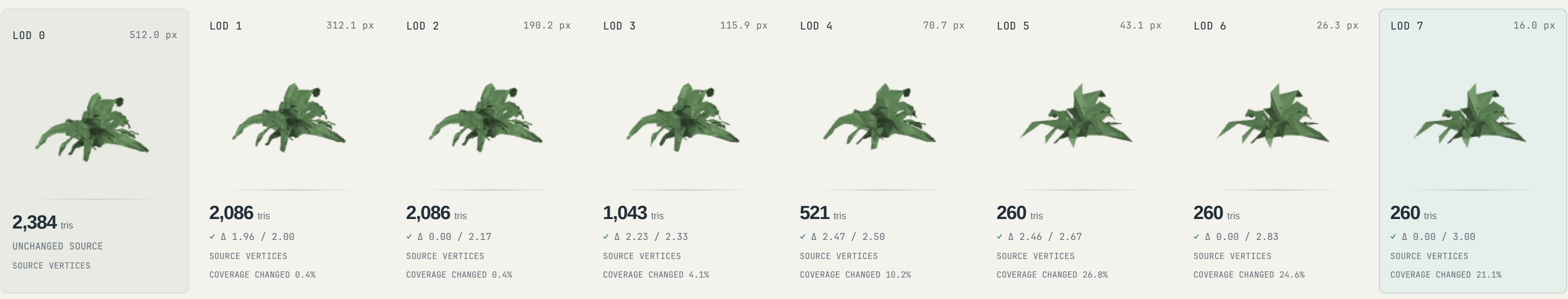
Fern fronds · candidate / shared

8 proposals per level; identical pixel limits. Geometry-only, without opacity or shading. · 5 runtime levels / 8 audit slots

Chain bake: 31.63 s · **Vertices: shared original buffer** · generation + audit

Dense tail check: passed (706 views)

−89.1% at LOD 7
2,384 → 260 tris



LOD 0 512.0 px	LOD 1 312.1 px	LOD 2 190.2 px	LOD 3 115.9 px	LOD 4 70.7 px	LOD 5 43.1 px	LOD 6 26.3 px	LOD 7 16.0 px
2,384 tris	2,086 tris	2,086 tris	1,043 tris	521 tris	260 tris	260 tris	260 tris
UNCHANGED SOURCE	✓ Δ 1.96 / 2.00	✓ Δ 0.00 / 2.17	✓ Δ 2.23 / 2.33	✓ Δ 2.47 / 2.50	✓ Δ 2.46 / 2.67	✓ Δ 0.00 / 2.83	✓ Δ 0.00 / 3.00
SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES
	COVERAGE CHANGED 0.4%	COVERAGE CHANGED 0.4%	COVERAGE CHANGED 4.1%	COVERAGE CHANGED 10.2%	COVERAGE CHANGED 26.8%	COVERAGE CHANGED 24.6%	COVERAGE CHANGED 21.1%

07 / NATURE · FERN

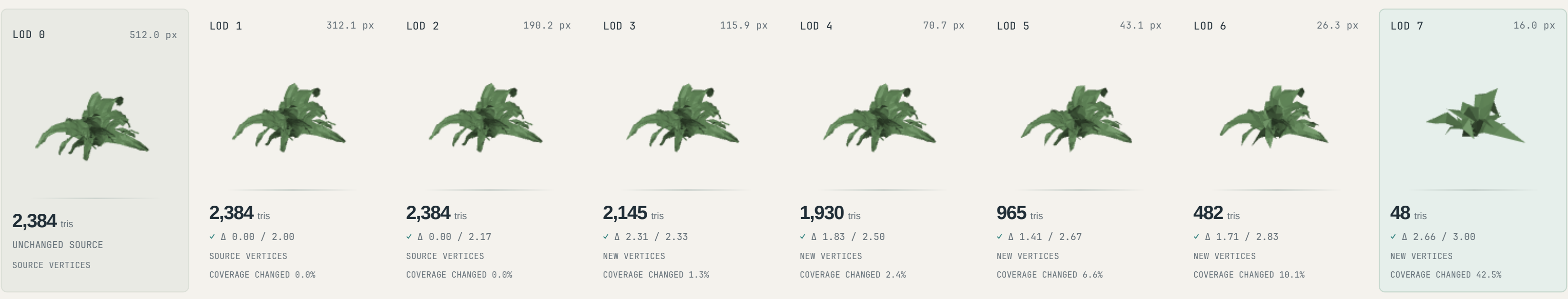
Fern fronds · before / rebuilt

8 proposals per level; identical pixel limits. Geometry-only, without opacity or shading. · 6 runtime levels / 8 audit slots

Chain bake: 3.75 s · **Vertices: mixed source + new buffers (rebuild mode)** · generation + audit

Dense tail check: passed (706 views)

−98.0% at LOD 7
2,384 → 48 tris



LOD 0 512.0 px	LOD 1 312.1 px	LOD 2 190.2 px	LOD 3 115.9 px	LOD 4 70.7 px	LOD 5 43.1 px	LOD 6 26.3 px	LOD 7 16.0 px
2,384 tris	2,384 tris	2,384 tris	2,145 tris	1,930 tris	965 tris	482 tris	48 tris
UNCHANGED SOURCE	✓ Δ 0.00 / 2.00	✓ Δ 0.00 / 2.17	✓ Δ 2.31 / 2.33	✓ Δ 1.83 / 2.50	✓ Δ 1.41 / 2.67	✓ Δ 1.71 / 2.83	✓ Δ 2.66 / 3.00
SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES
	COVERAGE CHANGED 0.0%	COVERAGE CHANGED 0.0%	COVERAGE CHANGED 1.3%	COVERAGE CHANGED 2.4%	COVERAGE CHANGED 6.6%	COVERAGE CHANGED 10.1%	COVERAGE CHANGED 42.5%

08 / NATURE · FERN

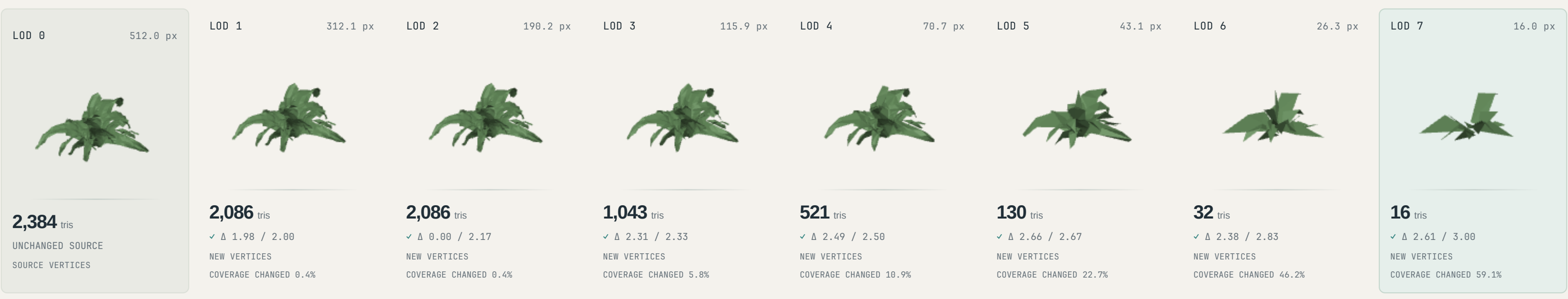
Fern fronds · candidate / rebuilt

8 proposals per level; identical pixel limits. Geometry-only, without opacity or shading. · 7 runtime levels / 8 audit slots

Chain bake: 31.26 s · **Vertices: rebuilt / new buffers** · generation + audit

Dense tail check: passed (706 views)

−99.3% at LOD 7
2,384 → 16 tris



LOD 0 512.0 px	LOD 1 312.1 px	LOD 2 190.2 px	LOD 3 115.9 px	LOD 4 70.7 px	LOD 5 43.1 px	LOD 6 26.3 px	LOD 7 16.0 px
2,384 tris	2,086 tris	2,086 tris	1,043 tris	521 tris	130 tris	32 tris	16 tris
UNCHANGED SOURCE	✓ Δ 1.98 / 2.00	✓ Δ 0.00 / 2.17	✓ Δ 2.31 / 2.33	✓ Δ 2.49 / 2.50	✓ Δ 2.66 / 2.67	✓ Δ 2.38 / 2.83	✓ Δ 2.61 / 3.00
SOURCE VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES
	COVERAGE CHANGED 0.4%	COVERAGE CHANGED 0.4%	COVERAGE CHANGED 5.8%	COVERAGE CHANGED 10.9%	COVERAGE CHANGED 22.7%	COVERAGE CHANGED 46.2%	COVERAGE CHANGED 59.1%

09 / NATURE · TREE

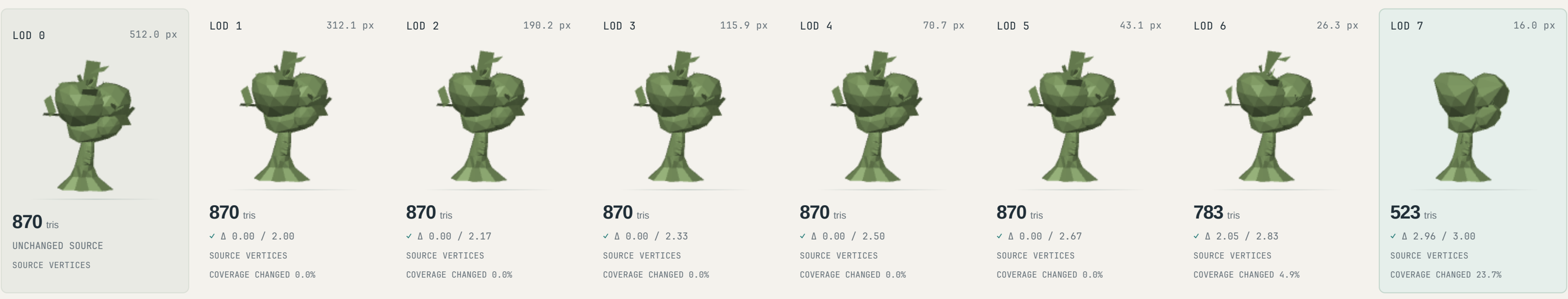
Broadleaf tree · before / shared

8 proposals per level; identical pixel limits. Geometry-only, without opacity or shading. · 3 runtime levels / 8 audit slots

Chain bake: 0.80 s · **Vertices: shared original buffer** · generation + audit

Dense tail check: passed (706 views)

−39.9% at LOD 7
870 → 523 tris



LOD 0 512.0 px	LOD 1 312.1 px	LOD 2 190.2 px	LOD 3 115.9 px	LOD 4 70.7 px	LOD 5 43.1 px	LOD 6 26.3 px	LOD 7 16.0 px
870 tris	870 tris	870 tris	870 tris	870 tris	870 tris	783 tris	523 tris
UNCHANGED SOURCE	✓ Δ 0.00 / 2.00	✓ Δ 0.00 / 2.17	✓ Δ 0.00 / 2.33	✓ Δ 0.00 / 2.50	✓ Δ 0.00 / 2.67	✓ Δ 2.05 / 2.83	✓ Δ 2.96 / 3.00
SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES
	COVERAGE CHANGED 0.0%	COVERAGE CHANGED 0.0%	COVERAGE CHANGED 0.0%	COVERAGE CHANGED 0.0%	COVERAGE CHANGED 0.0%	COVERAGE CHANGED 4.9%	COVERAGE CHANGED 23.7%

10 / NATURE · TREE

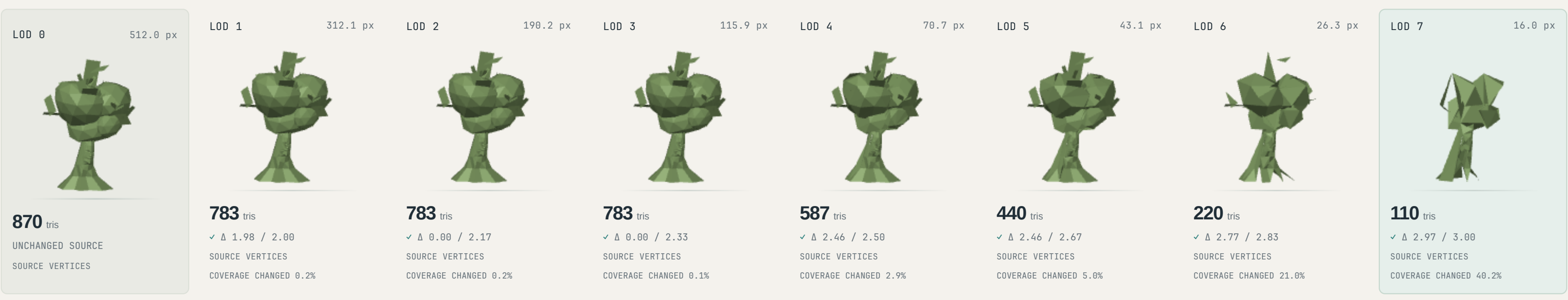
Broadleaf tree · candidate / shared

8 proposals per level; identical pixel limits. Geometry-only, without opacity or shading. · 6 runtime levels / 8 audit slots

Chain bake: 8.30 s · **Vertices: shared original buffer** · generation + audit

Dense tail check: passed (706 views)

−87.4% at LOD 7
870 → 110 tris



LOD 0 512.0 px	LOD 1 312.1 px	LOD 2 190.2 px	LOD 3 115.9 px	LOD 4 70.7 px	LOD 5 43.1 px	LOD 6 26.3 px	LOD 7 16.0 px
870 tris	783 tris	783 tris	783 tris	587 tris	440 tris	220 tris	110 tris
UNCHANGED SOURCE	✓ Δ 1.98 / 2.00	✓ Δ 0.00 / 2.17	✓ Δ 0.00 / 2.33	✓ Δ 2.46 / 2.50	✓ Δ 2.46 / 2.67	✓ Δ 2.77 / 2.83	✓ Δ 2.97 / 3.00
SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES
	COVERAGE CHANGED 0.2%	COVERAGE CHANGED 0.2%	COVERAGE CHANGED 0.1%	COVERAGE CHANGED 2.9%	COVERAGE CHANGED 5.0%	COVERAGE CHANGED 21.0%	COVERAGE CHANGED 40.2%

11 / NATURE · TREE

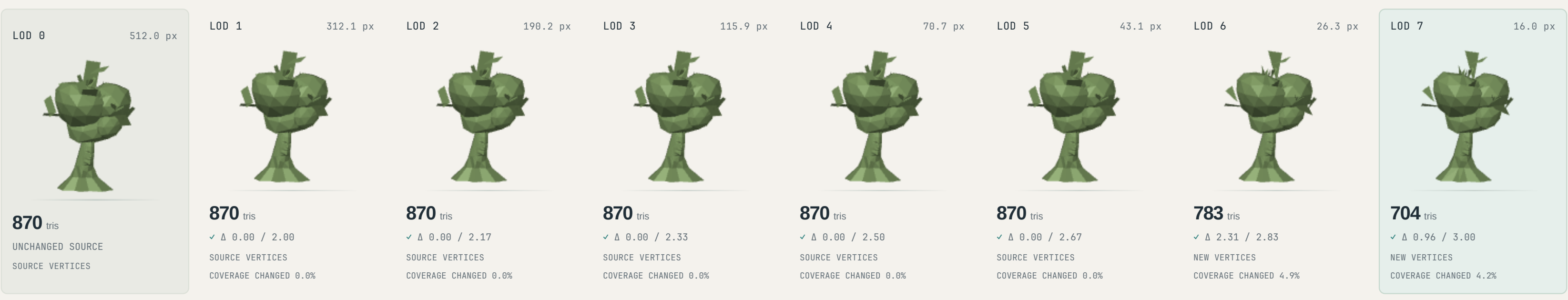
Broadleaf tree · before / rebuilt

8 proposals per level; identical pixel limits. Geometry-only, without opacity or shading. · 3 runtime levels / 8 audit slots

Chain bake: 0.73 s · **Vertices: mixed source + new buffers (rebuild mode)** · generation + audit

Dense tail check: passed (706 views)

−19.1% at LOD 7
870 → 704 tris



LOD 0 512.0 px	LOD 1 312.1 px	LOD 2 190.2 px	LOD 3 115.9 px	LOD 4 70.7 px	LOD 5 43.1 px	LOD 6 26.3 px	LOD 7 16.0 px
870 tris	870 tris	870 tris	870 tris	870 tris	870 tris	783 tris	704 tris
UNCHANGED SOURCE	✓ Δ 0.00 / 2.00	✓ Δ 0.00 / 2.17	✓ Δ 0.00 / 2.33	✓ Δ 0.00 / 2.50	✓ Δ 0.00 / 2.67	✓ Δ 2.31 / 2.83	✓ Δ 0.96 / 3.00
SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	NEW VERTICES	NEW VERTICES
	COVERAGE CHANGED 0.0%	COVERAGE CHANGED 0.0%	COVERAGE CHANGED 0.0%	COVERAGE CHANGED 0.0%	COVERAGE CHANGED 0.0%	COVERAGE CHANGED 4.9%	COVERAGE CHANGED 4.2%

12 / NATURE · TREE

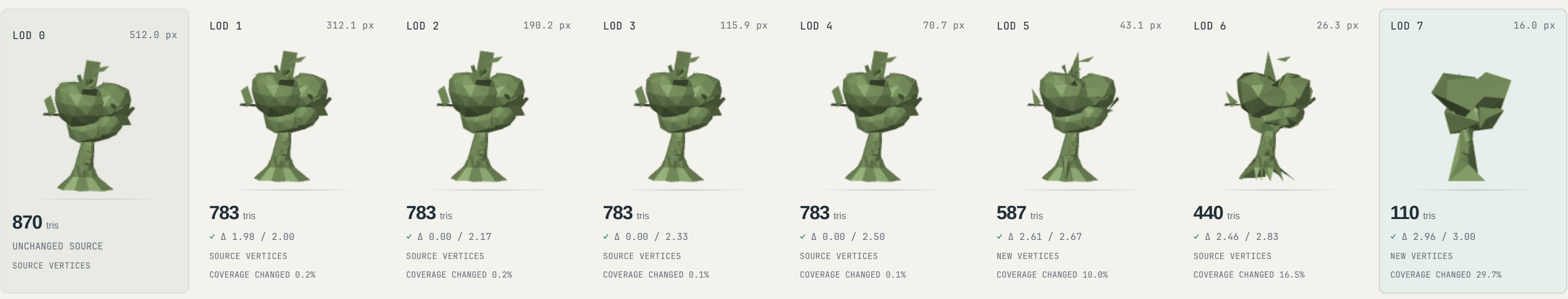
Broadleaf tree · candidate / rebuilt

8 proposals per level; identical pixel limits. Geometry-only, without opacity or shading. · 5 runtime levels / 8 audit slots

Chain bake: 9.91 s · **Vertices: mixed source + new buffers (rebuild mode)** · generation + audit

Dense tail check: passed (706 views)

−87.4% at LOD 7
870 → 110 tris



LOD 0 512.0 px	LOD 1 312.1 px	LOD 2 190.2 px	LOD 3 115.9 px	LOD 4 70.7 px	LOD 5 43.1 px	LOD 6 26.3 px	LOD 7 16.0 px
870 tris	783 tris	783 tris	783 tris	783 tris	587 tris	440 tris	110 tris
UNCHANGED SOURCE	✓ Δ 1.98 / 2.00	✓ Δ 0.00 / 2.17	✓ Δ 0.00 / 2.33	✓ Δ 0.00 / 2.50	✓ Δ 2.61 / 2.67	✓ Δ 2.46 / 2.83	✓ Δ 2.96 / 3.00
SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	NEW VERTICES	SOURCE VERTICES	NEW VERTICES
	COVERAGE CHANGED 0.2%	COVERAGE CHANGED 0.2%	COVERAGE CHANGED 0.1%	COVERAGE CHANGED 0.1%	COVERAGE CHANGED 10.0%	COVERAGE CHANGED 16.5%	COVERAGE CHANGED 29.7%

13 / NATURE · GRASS

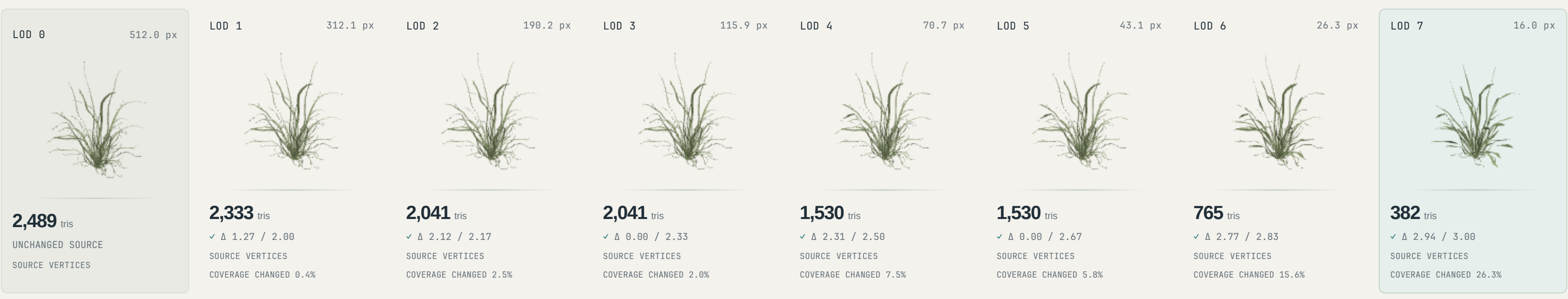
Grass clump · candidate / shared

8 proposals per level; identical pixel limits. Geometry-only, without opacity or shading. · 6 runtime levels / 8 audit slots

Chain bake: 13.76 s · **Vertices: shared original buffer** · generation + audit

Dense tail check: passed (706 views)

−84.7% at LOD 7
2,489 → 382 tris



LOD 0 512.0 px	LOD 1 312.1 px	LOD 2 190.2 px	LOD 3 115.9 px	LOD 4 70.7 px	LOD 5 43.1 px	LOD 6 26.3 px	LOD 7 16.0 px
2,489 tris	2,333 tris	2,041 tris	2,041 tris	1,530 tris	1,530 tris	765 tris	382 tris
UNCHANGED SOURCE	✓ Δ 1.27 / 2.00	✓ Δ 2.12 / 2.17	✓ Δ 0.00 / 2.33	✓ Δ 2.31 / 2.50	✓ Δ 0.00 / 2.67	✓ Δ 2.77 / 2.83	✓ Δ 2.94 / 3.00
SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES
	COVERAGE CHANGED 0.4%	COVERAGE CHANGED 2.5%	COVERAGE CHANGED 2.0%	COVERAGE CHANGED 7.5%	COVERAGE CHANGED 5.8%	COVERAGE CHANGED 15.6%	COVERAGE CHANGED 26.3%

14 / NATURE · GRASS

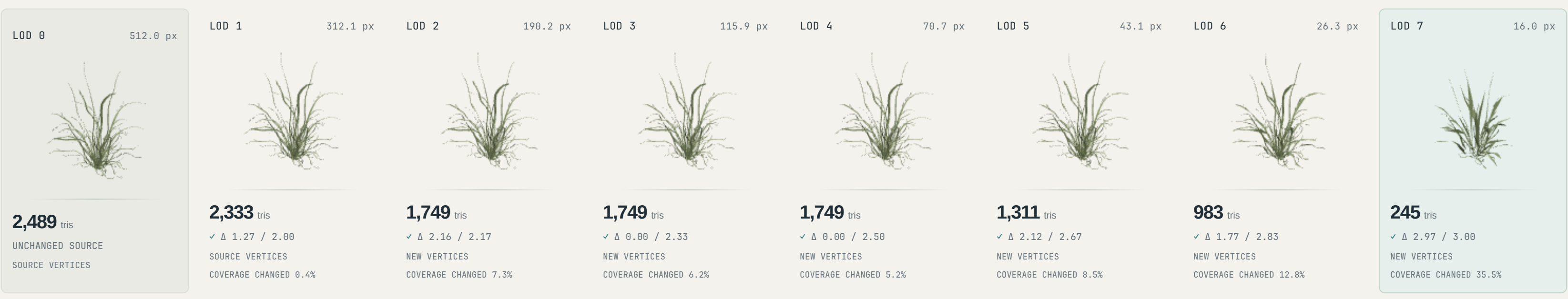
Grass clump · candidate / rebuilt

8 proposals per level; identical pixel limits. Geometry-only, without opacity or shading. · 6 runtime levels / 8 audit slots

Chain bake: 15.42 s · **Vertices: mixed source + new buffers (rebuild mode)** · generation + audit

Dense tail check: passed (706 views)

−90.2% at LOD 7
2,489 → 245 tris



LOD 0 512.0 px	LOD 1 312.1 px	LOD 2 190.2 px	LOD 3 115.9 px	LOD 4 70.7 px	LOD 5 43.1 px	LOD 6 26.3 px	LOD 7 16.0 px
2,489 tris	2,333 tris	1,749 tris	1,749 tris	1,749 tris	1,311 tris	983 tris	245 tris
UNCHANGED SOURCE	✓ Δ 1.27 / 2.00	✓ Δ 2.16 / 2.17	✓ Δ 0.00 / 2.33	✓ Δ 0.00 / 2.50	✓ Δ 2.12 / 2.67	✓ Δ 1.77 / 2.83	✓ Δ 2.97 / 3.00
SOURCE VERTICES	SOURCE VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES
	COVERAGE CHANGED 0.4%	COVERAGE CHANGED 7.3%	COVERAGE CHANGED 6.2%	COVERAGE CHANGED 5.2%	COVERAGE CHANGED 8.5%	COVERAGE CHANGED 12.8%	COVERAGE CHANGED 35.5%

15 / NATURE · SHRUB

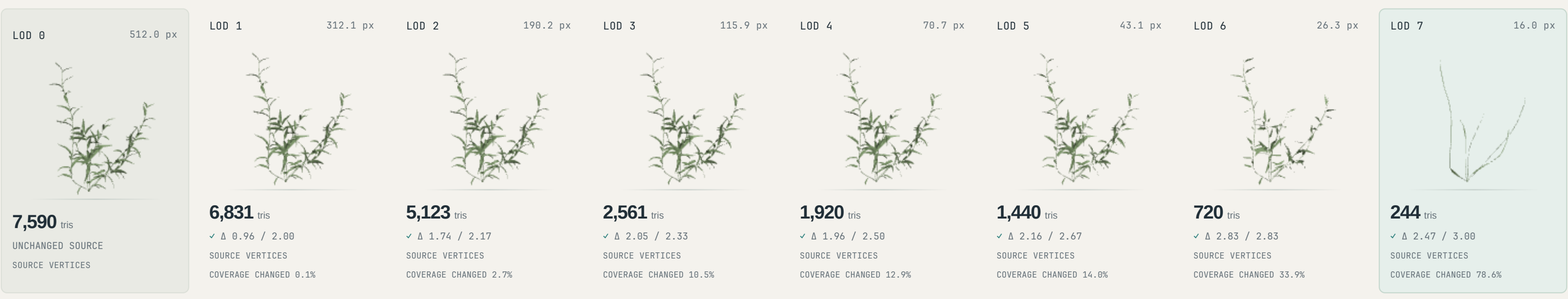
Leafy shrub · candidate / shared

8 proposals per level; identical pixel limits. Geometry-only, without opacity or shading. · 8 runtime levels / 8 audit slots

Chain bake: 28.89 s · **Vertices: shared original buffer** · generation + audit

Dense tail check: passed (706 views)

−96.8% at LOD 7
7,590 → 244 tris



LOD 0 512.0 px	LOD 1 312.1 px	LOD 2 190.2 px	LOD 3 115.9 px	LOD 4 70.7 px	LOD 5 43.1 px	LOD 6 26.3 px	LOD 7 16.0 px
7,590 tris	6,831 tris	5,123 tris	2,561 tris	1,920 tris	1,440 tris	720 tris	244 tris
UNCHANGED SOURCE	✓ Δ 0.96 / 2.00	✓ Δ 1.74 / 2.17	✓ Δ 2.05 / 2.33	✓ Δ 1.96 / 2.50	✓ Δ 2.16 / 2.67	✓ Δ 2.83 / 2.83	✓ Δ 2.47 / 3.00
SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES
	COVERAGE CHANGED 0.1%	COVERAGE CHANGED 2.7%	COVERAGE CHANGED 10.5%	COVERAGE CHANGED 12.9%	COVERAGE CHANGED 14.0%	COVERAGE CHANGED 33.9%	COVERAGE CHANGED 78.6%

16 / NATURE · SHRUB

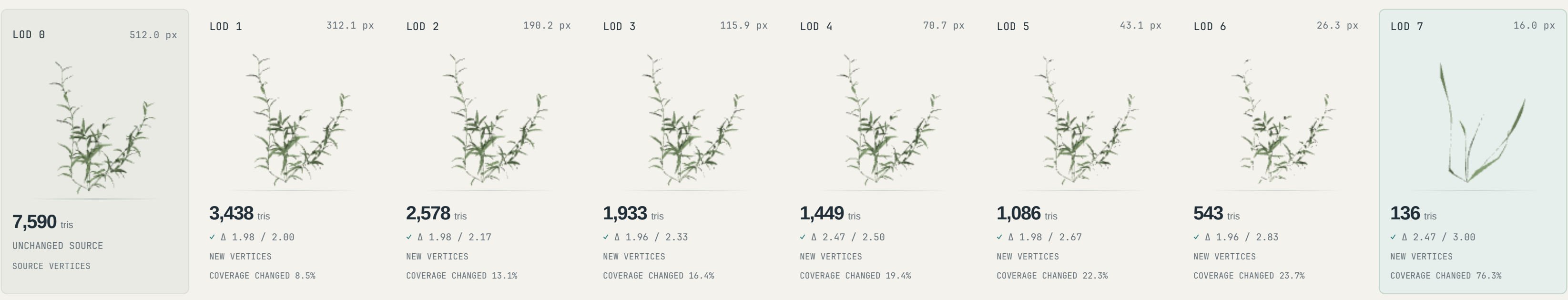
Leafy shrub · candidate / rebuilt

8 proposals per level; identical pixel limits. Geometry-only, without opacity or shading. · 8 runtime levels / 8 audit slots

Chain bake: 31.19 s · **Vertices: rebuilt / new buffers** · generation + audit

Dense tail check: passed (706 views)

−98.2% at LOD 7
7,590 → 136 tris



LOD 0 512.0 px	LOD 1 312.1 px	LOD 2 190.2 px	LOD 3 115.9 px	LOD 4 70.7 px	LOD 5 43.1 px	LOD 6 26.3 px	LOD 7 16.0 px
7,590 tris	3,438 tris	2,578 tris	1,933 tris	1,449 tris	1,086 tris	543 tris	136 tris
UNCHANGED SOURCE	✓ Δ 1.98 / 2.00	✓ Δ 1.98 / 2.17	✓ Δ 1.96 / 2.33	✓ Δ 2.47 / 2.50	✓ Δ 1.98 / 2.67	✓ Δ 1.96 / 2.83	✓ Δ 2.47 / 3.00
SOURCE VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES
	COVERAGE CHANGED 8.5%	COVERAGE CHANGED 13.1%	COVERAGE CHANGED 16.4%	COVERAGE CHANGED 19.4%	COVERAGE CHANGED 22.3%	COVERAGE CHANGED 23.7%	COVERAGE CHANGED 76.3%

17 / NATURE · FLOWERS

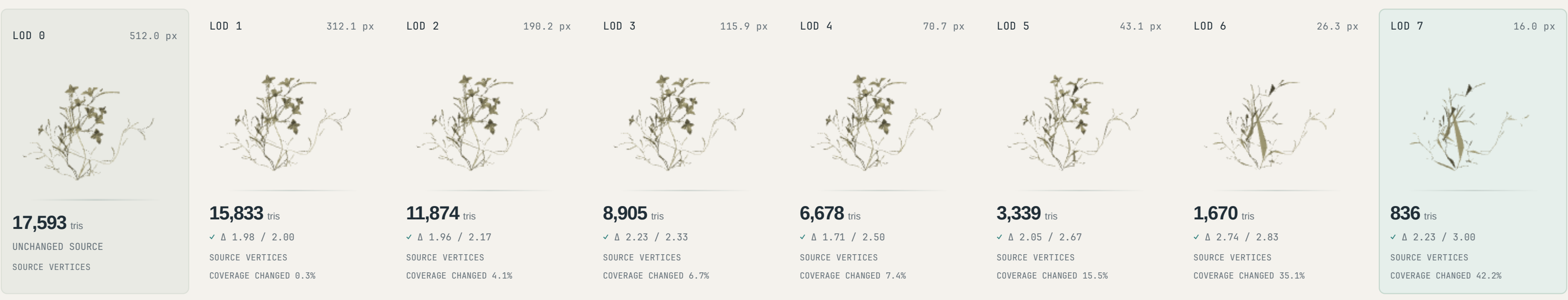
Flowering ground cover · candidate / shared

8 proposals per level; identical pixel limits. Geometry-only, without opacity or shading. · 8 runtime levels / 8 audit slots

Chain bake: 28.10 s · **Vertices: shared original buffer** · generation + audit

Dense tail check: passed (706 views)

−95.2% at LOD 7
17,593 → 836 tris



LOD 0 512.0 px	LOD 1 312.1 px	LOD 2 190.2 px	LOD 3 115.9 px	LOD 4 70.7 px	LOD 5 43.1 px	LOD 6 26.3 px	LOD 7 16.0 px
17,593 tris	15,833 tris	11,874 tris	8,905 tris	6,678 tris	3,339 tris	1,670 tris	836 tris
UNCHANGED SOURCE	✓ Δ 1.98 / 2.00	✓ Δ 1.96 / 2.17	✓ Δ 2.23 / 2.33	✓ Δ 1.71 / 2.50	✓ Δ 2.05 / 2.67	✓ Δ 2.74 / 2.83	✓ Δ 2.23 / 3.00
SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES
	COVERAGE CHANGED 0.3%	COVERAGE CHANGED 4.1%	COVERAGE CHANGED 6.7%	COVERAGE CHANGED 7.4%	COVERAGE CHANGED 15.5%	COVERAGE CHANGED 35.1%	COVERAGE CHANGED 42.2%

18 / NATURE · FLOWERS

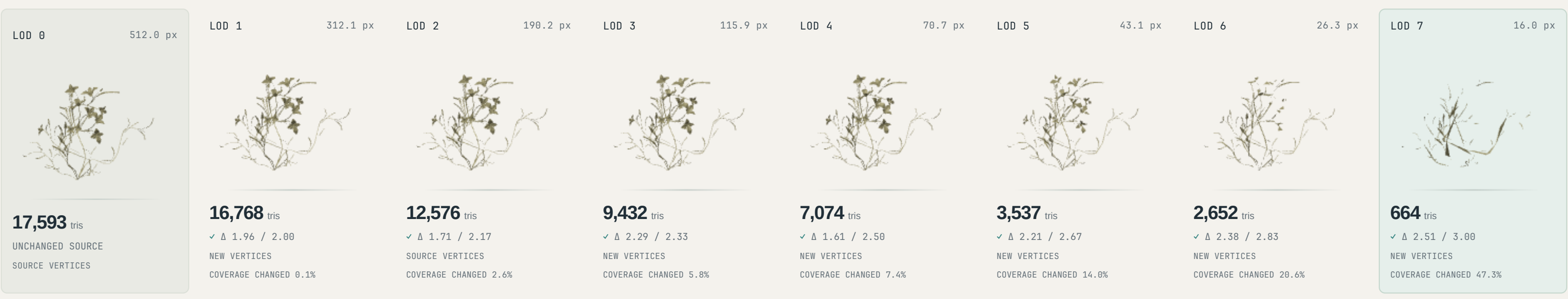
Flowering ground cover · candidate / rebuilt

8 proposals per level; identical pixel limits. Geometry-only, without opacity or shading. · 8 runtime levels / 8 audit slots

Chain bake: 31.91 s · **Vertices: mixed source + new buffers (rebuild mode)** · generation + audit

Dense tail check: passed (706 views)

−96.2% at LOD 7
17,593 → 664 tris



LOD 0 512.0 px	LOD 1 312.1 px	LOD 2 190.2 px	LOD 3 115.9 px	LOD 4 70.7 px	LOD 5 43.1 px	LOD 6 26.3 px	LOD 7 16.0 px
17,593 tris	16,768 tris	12,576 tris	9,432 tris	7,074 tris	3,537 tris	2,652 tris	664 tris
UNCHANGED SOURCE	✓ Δ 1.96 / 2.00	✓ Δ 1.71 / 2.17	✓ Δ 2.29 / 2.33	✓ Δ 1.61 / 2.50	✓ Δ 2.21 / 2.67	✓ Δ 2.38 / 2.83	✓ Δ 2.51 / 3.00
SOURCE VERTICES	NEW VERTICES	SOURCE VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES
	COVERAGE CHANGED 0.1%	COVERAGE CHANGED 2.6%	COVERAGE CHANGED 5.8%	COVERAGE CHANGED 7.4%	COVERAGE CHANGED 14.0%	COVERAGE CHANGED 20.6%	COVERAGE CHANGED 47.3%

19 / NATURE · BRANCH

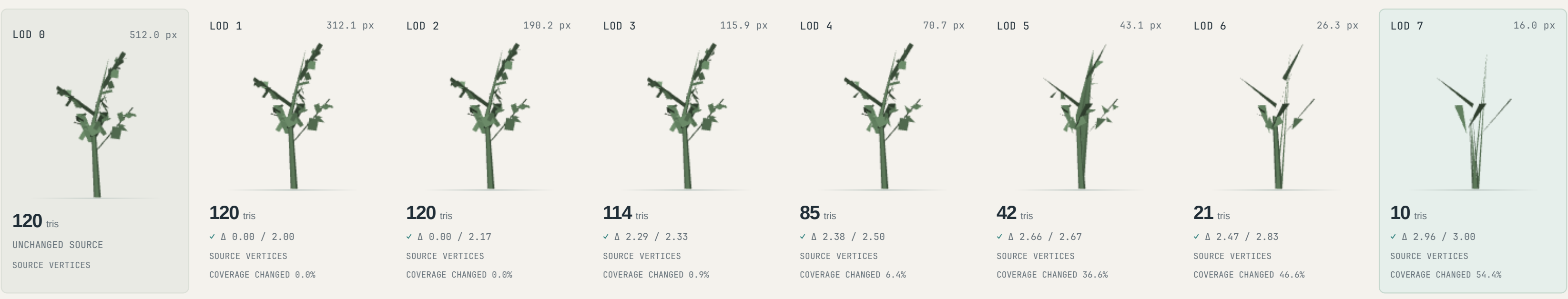
Leafy branch · candidate / shared

8 proposals per level; identical pixel limits. Geometry-only, without opacity or shading. · 6 runtime levels / 8 audit slots

Chain bake: 4.08 s · **Vertices: shared original buffer** · generation + audit

Dense tail check: passed (706 views)

−91.7% at LOD 7
120 → 10 tris



LOD 0 512.0 px	LOD 1 312.1 px	LOD 2 190.2 px	LOD 3 115.9 px	LOD 4 70.7 px	LOD 5 43.1 px	LOD 6 26.3 px	LOD 7 16.0 px
120 tris	120 tris	120 tris	114 tris	85 tris	42 tris	21 tris	10 tris
UNCHANGED SOURCE	✓ Δ 0.00 / 2.00	✓ Δ 0.00 / 2.17	✓ Δ 2.29 / 2.33	✓ Δ 2.38 / 2.50	✓ Δ 2.66 / 2.67	✓ Δ 2.47 / 2.83	✓ Δ 2.96 / 3.00
SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES	SOURCE VERTICES
	COVERAGE CHANGED 0.0%	COVERAGE CHANGED 0.0%	COVERAGE CHANGED 0.9%	COVERAGE CHANGED 6.4%	COVERAGE CHANGED 36.6%	COVERAGE CHANGED 46.6%	COVERAGE CHANGED 54.4%

20 / NATURE · BRANCH

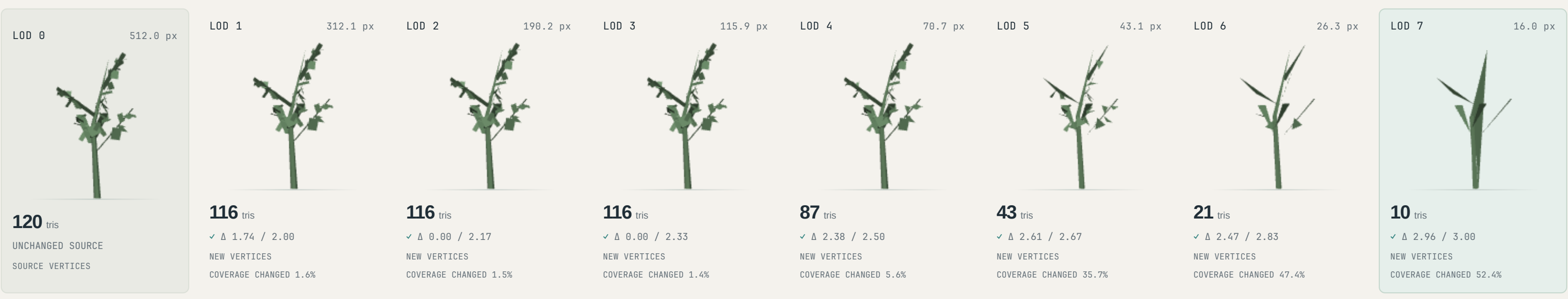
Leafy branch · candidate / rebuilt

8 proposals per level; identical pixel limits. Geometry-only, without opacity or shading. · 6 runtime levels / 8 audit slots

Chain bake: 8.35 s · **Vertices: rebuilt / new buffers** · generation + audit

Dense tail check: passed (706 views)

−91.7% at LOD 7
120 → 10 tris



LOD 0 512.0 px	LOD 1 312.1 px	LOD 2 190.2 px	LOD 3 115.9 px	LOD 4 70.7 px	LOD 5 43.1 px	LOD 6 26.3 px	LOD 7 16.0 px
120 tris	116 tris	116 tris	116 tris	87 tris	43 tris	21 tris	10 tris
UNCHANGED SOURCE	✓ Δ 1.74 / 2.00	✓ Δ 0.00 / 2.17	✓ Δ 0.00 / 2.33	✓ Δ 2.38 / 2.50	✓ Δ 2.61 / 2.67	✓ Δ 2.47 / 2.83	✓ Δ 2.96 / 3.00
SOURCE VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES	NEW VERTICES
	COVERAGE CHANGED 1.6%	COVERAGE CHANGED 1.5%	COVERAGE CHANGED 1.4%	COVERAGE CHANGED 5.6%	COVERAGE CHANGED 35.7%	COVERAGE CHANGED 47.4%	COVERAGE CHANGED 52.4%

Reading the tiles: Δ is the audited upper error against the previous LOD, followed by its pixel limit. Each reduced mesh also passed its direct-to-source limit. LOD 0 is unchanged.

Neutral clay uses geometric face normals, without textures. Presentation angles are not audit cameras. A low coverage error does not constrain shading.

THE VISUAL CONTRACT

512 → 16 px · 8 audit slots

2 → 3 px transitions; source cap 8 px. The cumulative policy reaches 6.89 px at the final level. Runtime meshes share exact duplicate slots.

THE AUDIT BEHIND THESE EXAMPLES

12 orthographic + 4 perspective views

Coverage is checked at 4× sampling with refinement to 8×. Search uses a separate camera set. These are finite sampled views, not a universal error guarantee.

WHAT REMAINS OUTSIDE THIS BOARD

Full-quality corpus audit

The default 642 + 64 camera preset has not been run across the full corpus. Texture images, opacity, skinning and morph targets are outside this version.

BLITZREMESHER / VEGETATION RESEARCH

C++20 · same pixel limits · both vertex modes

Actual meshes. CC0 source assets.

Moon rock   Quiver tree trunk   Metal stool   Bell X-1   Fern fronds · before / shared   Fern fronds · candidate / shared   Fern fronds · before / rebuilt   Fern fronds · candidate / rebuilt   Broadleaf tree · before / shared   Broadleaf tree · candidate / shared   Broadleaf tree · before / rebuilt   Broadleaf tree · candidate / rebuilt   Grass clump · candidate / shared   Grass clump · candidate / rebuilt   Leafy shrub · candidate / shared   Leafy shrub · candidate / rebuilt   Flowering ground cover · candidate / shared   Flowering ground cover · candidate / rebuilt   Leafy branch · candidate / shared   Leafy branch · candidate / rebuilt

20 measured examples; nature chains are unscored geometry-only demonstrations.